

General Problem Solving

What Are We Going to Do?

- Nothing “programming”
    - But will help at the end!
- Just trying to lay down the steps
    - Not autonomous thinking
    - Writing exact steps
- Understand the process and other things involved with the problem
- No special rules
    - Just testing how you do these activities

Instructions

- Form groups
  - 5 per each group
  - All girls can not be in the same group!
  - You got 3 minutes to form groups!

Remember This

- Input
  - Raw materials used to solve a problem
- Output
  - Result obtained after solving the problem
- Process
  - Converting input to output is the process
  - This has steps

Question 1

- Preparing a letter which can be posted
  - Identify the inputs
  - Identify the process
  - Identify the output
- Make assumptions if you may!
- Produce a group answer
  - Write down the answer in your books

Sample Answer - Question 1

- Input
  - A sheet of paper suitable to write the letter on and a pen
  - An envelope and a stamp
  - Glue
- Process
  1. Writing the letter
  2. Folding the letter and putting it into the envelope
  3. Pasting the envelope
  4. Writing the recipient's address on the envelope
  5. Sticking the stamp
- Output
  - A letter ready to be posted.

Question 2

- Dividing 40 pages and 80 pages books from a parcel of books between two siblings (say Judi gets 40 pages and Sam gets 80 pages)
    - Identify the inputs
    - Identify the process
    - Identify the output
- Make assumptions if you may!
- Produce a group answer
    - Write down the answer in your books

Sample Answer - Question 2

- Input
  - The parcel of books
- Process
  1. Opening the book parcel
  2. Taking a book out from the parcel
  3. If its 40 page, give it to Judi
  4. If its 80 page, give it to Sam
  5. Go to Step No. 2 till all the books are taken out of the parcel
- Output
  - Judi getting 40 pages books
  - Sam getting 80 pages books

Question 3

- Divide 100 toffees which are taken from a bag among 20 people.
    - Identify the inputs
    - Identify the process
    - Identify the output
- The bag will have only 100 toffees
- Produce a group answer
    - Write down the answer in your books

Sample Answer - Question 3

- Input
  - The parcel of toffees
- Process
  1. Opening the toffee bag
  2. Dived the value 100 by 20 (answer is 5)
  3. Count 5 toffees and take them out of the bag
  4. Give those 5 toffees to first person
  5. Repeat steps 3 and 4; for each and every other person
  6. Stop when all 20 people got 5 toffees
- Output
  - Every person got 5 toffees each

Alternate Thinking

- Imagine you come to school-by-school bus. If the bus breaks down on your way to school, you certainly will think of other alternative ways to **reach** school.
  - Don’t try to go home back!
- Note down the additional options you can think of!

Some Options

1. Coming to school by another school bus which goes to your school
2. If you have money, reaching the school by CTB or private bus
3. Walking to school along the road
4. Walking to school using a short-cut
5. If you have a way to inform your parents get their support to reach the school
6. Reaching the school by car or motor bike with the support of a trustworthy
7. Anything else …. Except skipping the school!

Redo Question 3

- Is it possible?
- Use alternate thinking!

Thank You…!!!